

AUTONOMOUS DATA SCIENTIST

AUTODS FINAL DATA SCIENCE REPORT

1. DATASET OVERVIEW

Rows : 119

Columns : 2

Column Names:

['Experience', 'Salary']

2. DATA QUALITY

Missing Values : 0

Duplicate Rows : 0

3. CLEANING

Original Rows : 120

Cleaned Rows : 119

Duplicates Removed : 1

4. EXPLORATORY DATA ANALYSIS

Numerical Columns:

['Experience', 'Salary']

Categorical Columns:

[]

Statistics:

{'Experience': {'mean': 5.789915966386555, 'median': 5.8, 'minimum': 1.1, 'maximum': 10.0, 'standard_deviation': 2.5406323434812275}, 'Salary': {'mean': 9.124033613445379, 'median': 8.25, 'minimum': 2.22, 'maximum': 19.87, 'standard_deviation': 5.169212856729988}}

Correlation:

{'Experience': {'Experience': 1.0, 'Salary': 0.9751}, 'Salary': {'Experience': 0.9751, 'Salary': 1.0}}

5. MODELING

Algorithm:

Linear Regression

Feature:

Experience

Target:

Salary

Training Rows:

95

Testing Rows:

24

MAE:

0.9914939878132228

RMSE:

1.1807813682562762

R2 Score:

0.9521878040458659

6. VISUALIZATION

Library:

Plotly

Plots Generated:

6

7. AI INSIGHTS

['The dataset contains no missing values.', 'No duplicate rows were found.', 'Correlation between Experience and Salary is 0.9751.', 'The first two numerical variables have a strong positive relationship.', 'A Linear Regression model was trained using the first numerical column to predict the second.', 'Plotly visualizations were generated for distributions, relationships, correlations and outliers.']

8. RECOMMENDATIONS

- Validate the model using unseen data.
- Perform cross-validation.
- Compare multiple machine learning algorithms.

- Add useful features.
- Monitor model performance.
- Use generated visualizations for analysis.

9. FINAL CONCLUSION

The AutoDS system successfully processed the dataset through profiling, cleaning, EDA, modeling, visualization and AI insight generation.

The generated visualizations provide graphical support for understanding distributions, relationships, correlations and possible outliers.

END OF AUTODS FINAL DATA SCIENCE REPORT